Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part a in this mark scheme extract

5 (a)	A1 A2 A3 A4	2.1.4	Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g. First 3 characters are upper case letters: Fff123456o Characters 4 to 9 are non-zero numbers: FFF101111o Consists of only 10 characters: FFF111111ee Final character correct for sum of numbers: FFF112233o		(4)

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part a in this mark scheme extract

5 (b) (i)	B1 B2	1.1.9	Award 1 mark each for any of: • Simple implementation (1) • Can be used for sorted or unsorted lists (1) • If the target is at the beginning of the data structure the search will be faster than a binary search	Accept disadvantages of a binary compared with a linear search.	(2)

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part a in this mark scheme extract

5 (b) (ii)	B3 B4	1.1.9	Award 1 mark each for any of: • May need to compare with all items in list before search complete (1) • May need longer time for searching a large list (1)	Accept advantages of a binary compared with a linear search.	(2)

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part a in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part a in this mark scheme extract

5 (c) (i)	C1 C2	2.6.2	Award 1 mark each up to a maximum of 2 for: • Both can use arguments / parameter passing (1) • Both can use local variables (1) • Both can be called from anywhere within the program (1) • Code can be reused without being rewritten (1) • Both can be library files (1) • The code can be independently tested for both (1)		(2)

Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part a in this mark scheme extract

5 (c) (ii)	C3 C4	2.6.2	Award two marks for a linked explanation such as: • A function must always return a result (1) whereas a procedure does not (1) • A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1) • The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1) • A function produces information (1) whereas a procedure performs a task (1)		(2)

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Type MS · Design validation checks for a code format

Answer 1 of 3 · Question 5 part a · 4 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part a in this mark scheme extract

Mark Scheme – Python Coding

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 1 of 10 · find part g in this mark scheme extract

1(a)	**The only correct answer is C** A is not correct because Boolean only has values True and False B is not correct because it only allows for one character D is not correct because it is used for mixed text and numeric data		(1)

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 2 of 10 · find part g in this mark scheme extract

1(b)	(an error that) produces incorrect /unintended output (1)	Accept an example of an explained logic error	(1)

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 3 of 10 · find part g in this mark scheme extract

1(d)	Any one from: 1. explain how the program works (1) 2. improve readability of the program (1) 3. make code more understandable (1)		(1)

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 4 of 10 · find part g in this mark scheme extract

1(e)	1. Global variables created in the main program / can be accessed from any part of the program / are in scope throughout the program (1) 2. Local variables are declared/created inside a function/blocked code (indented, brackets) and are only accessible there / are only in scope within the function in which they're declared (1)		(2)

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 5 of 10 · find part g in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 6 of 10 · find part g in this mark scheme extract

<table>
<tr><td>1(g)</td><td>For up to two tests
Award 1 mark for an appropriate validation test and 1 mark for an item of test data that would fail only the given test

<table><tr><th>Validation test</th><th>Erroneous data</th></tr><tr><td>Are the first two characters letters/upper case letters?</td><td>1234CCC</td></tr><tr><td>Are the third- and fourth-characters numerical digits?</td><td>CCCCBBB</td></tr><tr><td>Are the last three characters letters/upper case letters?</td><td>CC12345</td></tr></table></td><td>Each test should target a different feature of one of the requirements
Exactly 7 characters (in the stem)
First two characters are letters / upper case letters
Characters 3 & 4 are numeric
Last three characters are letters / upper case letters</td><td>(4)</td></tr>
</table>

Question	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 7 of 10 · find part g in this mark scheme extract

1(c)	**C#** 1. Semicolon at the end of line 9 [maxValue=5;] (1) 2. Consistent capitalisation of 'count' on lines 10, 11, 13 and 14 (1) 3. constantValue needs to be defined and given the value 7 / replaced by value 7 (1) **Java** 1. Semicolon at the end of line 8 [maxValue = 5;] (1) 2. Consistent capitalisation of 'count' on lines 7, 9, 11 and 12 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1) **Python** 1. Colon needed at end of line 7 [count, maxValue):] (1) 2. Consistent capitalisation of 'count' on line 3, 7, 8 and 9 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1)	Candidates are required to open the file Q01c in the code editor. Amended code should be saved as Q01cFINISHED. Line numbers may have been changed by corrections to code (e.g. insertion of constantValue definition)	(3)

Question	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 8 of 10 · find part g in this mark scheme extract

<table>
<tr>
<td>1(f)</td>
<td>Award 1 mark for each correct pair of relational operators and 1 mark for correct Boolean operator for each condition.

<table>
<tr><th>Condition</th><th>Coding</th></tr>
<tr><td rowspan="2">Shop income more than £5000 or assistant sold at least 10 pairs</td><td>shopIncome > 5000; assistantSales >= 10 (1)</td></tr>
<tr><td>correct OR operator (1)</td></tr>
<tr><td rowspan="2">Shop income at least £2000 and assistant sold at least 5 pairs</td><td>shopIncome >= 2000; assistantSales >= 5 (1)</td></tr>
<tr><td>correct AND operator (1)</td></tr>
</table>

Accept >9, >1999 and >4 for respective >= comparisons</td>
<td>Candidates are required to open the file Q01f in the code editor.
Amended code should be saved as Q01fFINISHED.
Do not penalise candidates who attempt more than the stated requirements.
Do not award Boolean operator mark for single | or & as these are not correct.</td>
<td>(4)</td>
</tr>
</table>

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 9 of 10 · find part g in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 2 of 3 · Question 1 part g · 4 marks

2019 JUN · Paper 02 · extract 10 of 10 · find part g in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Design validation checks for a code format

Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part a in this mark scheme extract

2(a)	A1	**The only correct answer is D** ***A*** *is not correct because abstraction is a technique used in computational thinking* ***B*** *is not correct because encryption is a security measure* ***C*** *is not correct because iteration is a programming construct*		**1**

Question	mp	Answer	Additional guidance	Mark
2(b)(i)		Award **one** mark for each correct cell:		
	B1 B2	real/double/float: (1) • pay rate/18.75 (1) • total pay/356.25 (1) • tax/0.2/71.25 (1) • income/285.00 (1) int/integer: (1) • hours/19 (1)	Example must match data type for B2 Award suitable example even if not taken from payslip Accept string and matching example e.g. Hours	**2**

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Type MS · Design validation checks for a code format

Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
2(b)(ii)		Award **one** mark for each correct cell:		
	B3	**Input:** • Pay Rate/18.75 (1) • Hours/19 (1)		
	B4	**Process**: • Pay Rate * Hours/18.75 * 19 (1) • Total Pay * Tax/356.25 * 0.2 (1) • Total Pay – Tax/356.25 – 71.25 (1)	Don't need output e.g. total pay, just the calculation part e.g. pay rate * hours: Total pay = pay rate * hours Accept 0.2 or 20% as part of calculation for tax Ok to mix types e.g. Total Pay * 20% Accept descriptions such as *multiply the pay rate by hours (hours etc could be taken from input box/ B3 mark)*	
	B5	**Output:** • Total Pay/356.25 (1) • Tax/71.25 (1) • Income/285 (1)		**3**

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Type MS · Design validation checks for a code format

Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
2(c)(i)	C1 C2	Award **one** mark for **one** correct cell: Award **two** marks for all **three** correct cells: • **Normal:** any value between 0-255 e.g. 125 • **Boundary:** 0/255 • **Erroneous:** any negative value/any value 256 or above	Accept -1/256 for boundary data Don't accept non-int types	**2**

Question	mp	Answer	Additional Guidance	Mark
2(c)(ii)		Award **one** mark for any of the following up to a maximum of **four** marks:		
	C3	Input changed to int before comparison (1)		**4**
	C4	Relational operator used correctly to check for valid input (both **>=**0 and **<**256 or equivalent) (1)	Allow: >0 and/or <255	
	C5	'and' logical operator used to combine relational tests (1)	or operator acceptable if candidate rearranges the selection statement and test works for correct code option – check if C4 is now awardable	
	C6	Calling of `getHex(userInput)` (1)	Call must be inside print ( ) Subprogram and argument must be functionally correct Ignore additional variables/strings in print statement	

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Type MS · Design validation checks for a code format

Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
37 # complete the line to convert the user input to integer
38 userInput = int(input("please enter a number between 0 and 255 \n"))
39
40 # complete the test to check the input is between 0 and 255
41 if (userInput >=0) and (userInput <256):
42     # complete the line to call getHex subprogram
43     print(getHex(userInput))
44 else:
45     print("number out of range")
```

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Type MS · Design validation checks for a code format

Answer 3 of 3 · Question 2 part a · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark